



Determinants of Mode Shift Toward Urban Bus Systems: A Structural Equation Modelling Approach

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Abstract: Encouraging travellers to shift from private vehicles to public transport remains a persistent challenge in many urban areas, particularly where bus systems struggle to compete with more flexible modes. This study examines how different dimensions of service quality influence the intention to shift toward urban bus systems. A survey of 650 respondents was conducted in Yogyakarta, Indonesia, focusing on individuals currently relying on private or on-demand transport. A structural equation modelling approach was used to analyse the relationships between service attributes and behavioural intention. The results indicate that all service quality dimensions considered have a significant effect on mode shift intention, though their relative importance differs. Interpersonal aspects of service—particularly empathy and responsiveness—emerged as the strongest predictors, suggesting that user experience is shaped not only by operational performance but also by how passengers are treated. Reliability and tangible service features also contributed meaningfully, while accessibility and assurance played a more limited role. The model explained a substantial portion of the variance in behavioural intention, with an R^2 value exceeding 0.60. These findings point to the need for a more user-oriented approach in public transport planning. Improving operational performance alone may not be enough; how passengers experience service interactions matters just as much in shaping travel behaviour. For bus-based systems to become more competitive, attention to both service reliability and interpersonal quality appears essential. The study provides empirical support for strategies aimed at encouraging a sustained shift away from private transport in urban settings.

Keywords: Trans Jogja bus; Service quality; Willingness to shift; Partial least squares-structural equation modeling

1 Introduction

Trans Jogja Bus was developed as a mass transportation solution that integrates major routes in Yogyakarta to reduce dependence on private vehicles and ease urban congestion. However, in recent years, there have been indications that Trans Jogja Bus has not become the mode of choice for most residents of Yogyakarta. Research on youth groups shows low frequency of public transportation use, with most respondents rarely or never using public transportation services such as the Trans Jogja Bus despite the availability of such services. These findings indicate that the existence of a network alone is not sufficient to change people’s mobility habits [1].

This situation is further exacerbated by operational factors and user preferences. Several local studies evaluating the performance and effectiveness of Trans Jogja have identified key issues such as suboptimal service coverage and bus stop locations, inadequate service frequency and reliability, fleet condition and comfort, and the relative utility of buses compared to private vehicles. In addition, the emergence of ride-hailing and online motorcycle services provides a more flexible and often faster door-to-door alternative, thereby attracting potential users from public to private/paid modes of transportation [1–3].

The urgency of this research stems from pragmatic and policy needs. To design effective interventions, the government and operators need empirical evidence on which service attributes are the most hindering and which have the most potential to encourage mode shift. Improvements in services such as technical aspects (route networks, frequency, route priority), operational aspects (punctuality and fleet capacity), and user experience will be more effective if based on an analysis that measures the extent to which each factor contributes to the desire to shift. Other

factors that also influence the use of Bus Trans Jogja are passenger perceptions, such as safety, security, comfort, equity, affordability, and regularity [4]. Therefore, research that examines the effect of service quality on the intention to shift to using the Trans Jogja Bus is very important for formulating realistic and effective urban transportation policies [5, 6].

Although several previous studies have highlighted the importance of improving service quality in attracting public interest to shift to using Trans Jogja Bus, most of them are still limited to descriptive analyses of user satisfaction or service operational performance. Studies that empirically measure the extent to which each dimension of service quality, such as reliability, responsiveness, assurance, empathy, tangibles, and accessibility, influences the public's intention to shift from private vehicles or online transportation to the Trans Jogja Bus are still very limited. Furthermore, most previous studies have not used the partial least squares-structural equation modeling (PLS-SEM) approach to comprehensively test the causal relationships between constructs in the context of urban transport in medium-sized cities such as Yogyakarta.

Therefore, this study attempts to fill this gap by developing an empirical model that can quantitatively explain how service quality dimensions influence people's intention to shift to using the Trans Jogja Bus. By identifying the services that most influence the decision to shift modes of transport, the results of this study are expected not only to contribute theoretically to the development of service quality-based transport behaviour models but also to provide practical input for local governments and operators in formulating strategies to improve the competitiveness of public transport in urban areas such as Yogyakarta.

2 Literature Review

2.1 Public Transportation

Public transportation is a mass transit system that includes modes such as buses, BRT, light rail, and ferries designed to provide reliable, efficient, and affordable mobility for urban residents. This system is considered a key component of urban transportation infrastructure because it offers high transport capacity with lower energy consumption and emissions per passenger compared to private vehicles [7]. The main objectives are to reduce dependence on private vehicles, ease traffic congestion, reduce air pollution, and provide inclusive transportation access, especially for vulnerable groups such as the elderly and people with disabilities [7, 8].

The benefits of public transportation are diverse, ranging from environmental and economic to social aspects. For example, in the Transit Oriented Development (TOD) concept, the use of public transportation can reduce greenhouse gas emissions and air pollution and reduce the need for large parking areas, thus supporting the development of greener and denser cities [9]. On the public health side, access to these public services increases active mobility (such as walking to bus stops), which has a positive impact on the physical and mental well-being of residents. In addition, public transportation systems create socioeconomic opportunities through mobility inclusion for all demographic groups, stimulate integrated land use, and strengthen the city's resilience to density challenges and environmental risks [10].

2.2 Factors influencing the Desire to Shift to Public Transportation

Currently, the use of private vehicles in urban areas is quite high. This has caused new problems for urban transportation, such as traffic congestion, air pollution, an increase in accidents, and insufficient parking spaces. On the other hand, the high use of private vehicles can reduce the existence of public transportation. However, amid the high demand for private vehicles, there is actually an opportunity to shift to public transportation if the service is made more competitive [11]. Although private vehicles offer advantages in terms of frequency and reliability, if public transportation services offer competitive fares, accessibility, and good service quality, accompanied by "push-pull" intervention policies, this could trigger a shift to public transportation [12]. Several studies have been conducted that have successfully revealed many factors that can influence people to shift to using public transportation. These factors can be seen in Table 1.

Based on various previous studies, as shown in Table 1, it can be seen that there are many factors that influence the intention to shift to public transport. However, the approaches and results obtained show significant differences depending on the context and analysis model used. Some emphasise the importance of operational aspects such as service frequency and punctuality as the main factors for mode shift [13, 14], while others highlight intermodal integration and payment systems as more decisive factors in the transition to public transport [15, 16]. These differences reflect variations in causal logic between technical attributes and user perceptions. Furthermore, studies focusing on cities with established transport networks tend to show that comfort and travel information variables have a dominant effect on the intention to shift [17], while research in developing countries shows that cost and physical accessibility factors have a more direct influence on the decision to shift [4, 18].

Theoretically, the relationship between variables also shows causal complexity that needs to be considered. For example, the variables of price and travel time not only have a direct effect on the intention to shift, but also indirectly through perceptions of comfort, reliability, and safety [19]. On the other hand, there are studies that show that built

environment factors and risk perceptions are important mediators that link service availability to the decision to shift modes [20, 21]. These findings confirm that a comprehensive conceptual model needs to consider a combination of functional, psychological, and spatial factors to explain the intention to shift more accurately.

Table 1. Factors influencing the desire to shift to public transportation

No.	Factors Influencing the Shift to Public Transport	Reference
1	Service frequency & headway regularity	[13]
2	Total travel time (in vehicle + waiting + transfers)	[22]
3	Integrated ticketing & easy payment	[15]
4	Fare level & price sensitivity	[23]
5	First/last mile connectivity	[24]
6	Real-time information & wayfinding	[17]
7	Accessibility & onboard comfort	[25]
8	Demand-management/parking policies	[18]
9	Integrating shared micromobility	[16]
10	New BRT services	[26]
11	Built environment & service quality	[20]
12	Risk perception and convenience	[21]
13	Perceived health risk	[27]
14	Security/risk sensitivity priority	[19]
15	Service reliability/on-time performance	[14]

2.3 Partial Least Squares-Structural Equation Modeling

PLS-SEM is an alternative method to covariance-based structural equation modeling (CB-SEM), which has historically been more commonly used when analyzing data using structural equation modeling (SEM) [28]. PLS-SEM is a variance-based SEM method that focuses on maximizing the variance explained in endogenous constructs and is designed for prediction and theory development purposes in addition to theoretical confirmation. This methodology is particularly suitable for studies with complex models, non-normal data, small sample sizes, and a combination of reflective and formative indicators, making it adaptable to modern transportation research and social science conditions [28, 29]. The PLS-SEM algorithm estimates latent construct scores through simple iterations starting from indicators, then constructs, back to indicators until convergence, with the practical advantage that the normality assumption is not as strict as in CB-SEM.

The main purpose of using PLS-SEM is to estimate causal relationships between latent constructs with a focus on prediction and theory development, especially when the model is complex, and the data does not fully meet the assumption of normality, making this method more flexible than covariance-based SEM [30]. In addition, PLS-SEM makes an important contribution to the evaluation of measurement model validity, particularly through the confirmatory composite analysis (CCA) approach, which allows researchers to test the quality of composite measurements and ensure the reliability and validity of constructs before proceeding to the structural model testing stage [31]. PLS-SEM is also designed to estimate relationships between latent constructs with a primary focus on predictive accuracy and theory development, making it a superior method when research models are complex, data do not meet normality assumptions, or sample sizes are limited [32].

Furthermore, PLS-SEM is not only used to test the significance of relationships, but also to assess out-of-sample predictive ability through techniques such as PLSpredict, so that the results of the analysis are more relevant for practical decision-making and evidence-based policy [33]. Thus, PLS-SEM is a very useful tool in social, business, and transportation research, as it combines the goals of theory development with predictive application needs.

The analysis was conducted in two stages of systematic testing, namely a measurement model to ensure that latent constructs were measured reliably and validly, and a structural model to test the relationships between latent constructs and estimate the strength of causal paths (path coefficients) and the predictive effectiveness of the model [34]. The stages of PLS-SEM testing can be seen in Table 2 and Table 3.

2.4 Implementation of Partial Least Squares-Structural Equation Modeling (PLS-SEM) in the Transportation Sector

PLS-SEM is increasingly being implemented in transportation research due to its ability to analyze complex relationships between latent factors, such as service quality, user satisfaction, loyalty, and mode shifting intentions. This method is considered suitable for analyzing data with relatively small sample sizes, non-normal data distributions, and predictive models. In the context of transportation, PLS-SEM has been used to evaluate factors

that influence passenger satisfaction, the adoption of smart transportation, and users' willingness to shift to public transportation modes, thereby providing a strong empirical basis for formulating policies and strategies to improve services.

Table 2. Measurement model testing

No.	Testing	Parameter	Threshold Value	Reference
1	Indicator reliability (reflective indicator)	Outer loadings (standardized indicator loadings)	≥ 0.70 (ideal), ≥ 0.50 (still accepted with caution)	[31]
2	Internal consistency reliability	Composite reliability (CR), ρ_A (Cronbach's α)	$CR \geq 0.70$	[31, 35]
3	Convergent validity	Average variance extracted (AVE)	$AVE \geq 0.50$	[35–37]
4	Discriminant validity	Heterotrait-monotrait ratio (HTMT, main), Fornell–Larcker/cross-loadings (complementary)	HTMT < 0.90 (conservative: <0.85 for very similar constructs)	[30]
5	Formative indicators (optional)	Indicator multicollinearity (VIF); significant weights	Variation inflation factor (VIF) < 5 (ideal <3); significant weight ($p < 0.05$)	[30, 38]

Table 3. Structural model testing

No.	Testing	Parameter	Threshold Value	Reference
1	Collinearity (inner VIF)	Variation inflation factor (VIF)	$VIF < 5$	[31, 38]
2	Path coefficients—significance	Path β ; bootstrap significance	Bootstrap $\geq 5,000$ resamples, p -value < 0.05	[30, 38]
3	Explanatory power	Coefficient of determination R^2	$R^2 \approx 0.75$ (substantial), 0.50 (moderate), 0.25 (weak)	[31, 38]
4	Effect size	f^2 (contribution of each predictor to R^2)	$f^2 = 0.02$ (small), 0.15 (medium), 0.35 (large)	[31]
5	Predictive relevance (in-sample)	Q^2 (blindfolding/redundancy)	$Q^2 > 0 \rightarrow$ model has predictive relevance	[38]
6	Out-of-sample predictive assessment	PLSpredict (RMSE/MAE vs benchmark)	Model is better if PLS error $\leq$ benchmark error (RMSE/MAE)	[33]
7	Robustness checks/reporting	Report t , p , CI, sample size, bootstrap method, indicator decision	Follow CCA/PLS-SEM reporting guidelines (report reliability, validity, R^2 , f^2 , Q^2 , PLSpredict)	[31]

Note: RMSE: root mean square error; MAE: mean absolute error; CCA: confirmatory composite analysis; PLS-SEM: partial least squares-structural equation modeling.

Several studies in the field of transportation that use PLS-SEM include testing the dimensions of public transportation service quality that affect passenger satisfaction [39], modeling the intention to adopt sustainable modes of transportation (walking, cycling, public transportation) using a combined norm activation model and theory of planned behavior (NAM-TPB) model [40]. Assessing the factors that determine the intention to adopt Connected and Autonomous Vehicles [41], mapping the mechanisms of factors that influence the severity of injuries in bus accidents [42], assessing the satisfaction and preferences of ride-hailing users as part of the urban transportation ecosystem [43], analyzing how the complexity of travel chains (combination of modes, transfers, distance) affects mode choice (including buses), as well as the willingness to use public transportation [44], developing a comprehensive framework (combining NAM-TPB + perception factors) to predict mode choice [45], identifying barriers to the adoption of sustainable transport (including public transport) in the context of African cities [46], testing how perceptions of service quality and safety contribute to bus passenger loyalty [47], analysing service quality factors (reliability, tangibles, assurance, responsiveness, empathy) that influence user satisfaction with the

Bandung Trans Metro Bus [48], analysing how cultural, social, psychological and personality factors influence bus mode choices [49] identifying constructs that influence the acceptance of electric shuttle bus systems [50], and identify factors that influence bus terminal user satisfaction, including mode facilities, zones, and movement patterns [51].

3 Methodology

3.1 Research Location

This research was conducted in the Special Region of Yogyakarta, focusing on three cities/districts served by the Trans Jogja bus route, as shown in Figure 1, namely Yogyakarta City, Sleman Regency, and Bantul Regency.



Figure 1. Research location map

3.2 Questionnaire Design

The research questionnaire was designed to be as effective as possible to facilitate respondents in answering questions. The questionnaire consisted of items on public transport service quality comprising six latent variables (reliability, responsiveness, assurance, empathy, tangibility, and accessibility) as exogenous variables, and willingness to shift to using Trans Jogja Bus as an endogenous variable. Each latent variable has question indicators with answer options using a 1–5 Likert scale based on their level of importance. This method was chosen because it provides a measurable picture of respondents' perceptions of service quality dimensions, such as reliability, assurance, responsiveness, empathy, tangibles, and accessibility. The service quality variables and indicators that make up each latent variable are presented in Table 4.

The indicators and statement items, as shown in Table 4, were adopted from the service quality dimension model developed in a study entitled “SERVQUAL: A Multiple-Item Scale for Measuring Consumer Perceptions of Service Quality” [52], in which the model developed five dimensions of service quality consisting of reliability, assurance, responsiveness, empathy, and tangibles. In this study, the service quality dimensions were supplemented with accessibility to suit the context of public transport in Indonesia, where ease of access to bus stops, ease of payment, and location affordability are important factors in service quality perceptions.

In addition, the items used in the questionnaire underwent cultural adaptation and local validation to ensure they were appropriate to the social context, language, and customs of Indonesian society. The adaptation process involved forward translation and back translation, followed by expert judgement from academics specialising in transportation to ensure equivalence of meaning and contextual relevance. Furthermore, validity and reliability tests were conducted using local respondent trial data to ensure that each statement item could be understood properly and was able to represent its original construct in the Indonesian context.

3.3 Data Collection

Data collection for this study was conducted by distributing questionnaires to travellers who did not use the Trans Jogja bus service but had the potential to shift to using it. The questionnaire consisted of three sections: the first section contained sociodemographic information about the respondents (gender, age, education level, and income); the second section contained information about the respondents' travel characteristics (vehicle used, distance travelled from origin to destination, travel time, and travel costs); and the third section contained information about factors that could influence a shift to Trans Jogja Bus. The collection and completion of the questionnaire was conducted offline and online via Google Forms over a period of two months (May–June 2024). The respondents selected as samples were from groups of people who use private vehicles (cars and motorcycles) and online transportation (online cars and online motorcycles) to support their daily activities.

Table 4. Variables and indicators

Variable	Symbol	Indicator
Reliability (REL)	REL1	Bus waiting time
	REL2	Bus travel time
	REL3	Bus frequency and arrival schedule
Responsiveness (RES)	RES1	Staff readiness
	RES2	Staff response to issues
	RES3	Speed in resolving issues
Assurance (ASS)	ASS1	Service guarantees
	ASS2	Passenger capacity
	ASS3	Comfort factors
Empathy (EMP)	EMP1	Information attitude
	EMP2	Service attitude
	EMP3	Friendliness
Tangible (TGL)	TGL1	Number/location of stops
	TGL2	Dedicated bus lane
	TGL3	Parking facilities
Accessibility (ACC)	ACC1	Ticket purchase ease
	ACC2	Route coverage
	ACC3	Ease of transfer
Willingness to shift (WTS)	WTS1	Intention to use
	WTS2	Likelihood of choosing bus
	WTS3	Primary travel choice

The sampling was conducted using purposive sampling, considering that this group was the potential target of the modal shift policy. A total of 650 respondents were successfully collected according to the predetermined criteria. Sampling was conducted in urban areas traversed by the Trans Jogja Bus route, such as educational centres (schools and campuses), shopping centres (markets or malls), health facilities, office centres, and other activity centres. This approach was taken to ensure that respondents had the potential or opportunity to use the Trans Jogja Bus in their daily travel activities.

3.4 Analysis and Hypotheses

The analytical method used in this study is PLS-SEM. The purpose of this analysis is to determine the extent of the influence of public transport service quality (reliability, responsiveness, assurance, empathy, tangibles, and accessibility) on passengers' willingness to shift to using the Trans Jogja Bus. This approach is not only descriptive in nature, describing the conditions in the field, but also explanatory, as it focuses on testing the relationship between variables and the extent of the influence of service quality on the desire to shift. Thus, this study is able to provide a deeper understanding of the factors that encourage users to shift to the Trans Jogja Bus. Based on these objectives, the hypotheses developed in this study are as follows:

H1. The reliability factor has a positive effect on passengers' willingness to shift to using the Trans Jogja bus.

H2. The responsiveness factor has a positive effect on passengers' willingness to shift to using the Trans Jogja bus.

H3. The assurance factor has a positive effect on passengers' willingness to shift to using the Trans Jogja Bus.

H4. The empathy factor has a positive effect on passengers' willingness to shift to using the Trans Jogja Bus.

H5. The tangible factor has a positive effect on passengers' willingness to shift to using the Trans Jogja Bus.

H6. The accessibility factor has a positive effect on passengers' willingness to shift to using the Trans Jogja Bus.

4 Result

4.1 Respondent Characteristics

The respondents collected in this study were private vehicle and online transportation users in the city of Yogyakarta. Based on the survey results, there were 650 respondents with the characteristics shown in Table 5.

4.2 Measurement Model Assessment

The measurement model assessment conducted in this study included convergent validity, discriminant validity, and reliability testing. The testing process was carried out using Smart PLS 4 software. The test results can be seen in Figure 2 and Table 6.

4.2.1 Convergent validity

Convergent validity aims to ensure that the indicators (manifest variables) used are truly capable of representing the same construct. In other words, indicators that are assumed to measure a particular dimension/latent variable must have a sufficiently strong correlation with each other. Convergent validity is measured using several parameters, including factor loadings (outer loadings), where the loading value of each indicator must be >0.7 [31], and average variance extracted (AVE) must be >0.5 [35–37], and composite reliability (CR) > 0.7 [35].

Table 5. Respondent characteristics

Respondent Characteristics	Category	Frequency	Percentage (%)
Gender	Male	249	38.31
	Female	401	61.69
Age	<24 years old	245	37.69
	24–44 years old	369	56.77
	>44 years old	36	5.54
Education	Primary and secondary education	213	32.77
	Diploma/Bachelor	363	55.85
	Masters/Doctorate	74	11.38
	Government employees	62	9.54
Employment status	State-owned company employees	39	6.00
	Army/Police	56	8.62
	Private employees	168	25.85
	Entrepreneur	69	10.62
	Housewife	53	8.15
	Student	185	28.46
	Other	18	2.77
Monthly income	<2,500,000 IDR	246	37.85
	2,500,000–4,999,999 IDR	159	24.46
	5,000,000–7,499,999 IDR	134	20.62
	>7,500,000 IDR	111	17.08
Vehicles used daily	Car	131	20.15
	Motor	341	52.46
	Online car	73	11.23
	Online motor	105	16.15

Based on the test results as shown in Figure 2 and Table 6, it can be seen that the outer loading values of all indicators are >0.7 , the AVE values of all variables are >0.5 , and the CR (ρ_A) and CR (ρ_c) values are >0.7 . These results indicate that all convergent validity requirements have been met.

4.2.2 Discriminant validity

Discriminant validity testing in this study used the Fornell-Larcker Criterion and Cross Loading methods. Discriminant validity through Fornell–Larcker ensures that each construct is more closely connected to its own indicators than to other constructs. This confirms that the square root of the AVE of each construct must be greater than its correlation with other constructs. Meanwhile, the discriminant validity test using cross-loadings ensures that each indicator has the highest loading on the intended construct, not on other constructs. The results of the discriminant validity test are presented in Table 7 and Table 8.

Based on the test results as shown in Table 7 and Table 8, it can be seen that the square value of each construct (AVE) is greater than its correlation with other constructs. Similarly, the cross-loading of each indicator with its construct has the highest value compared to its relationship with other constructs. These results indicate that the discriminant validity requirements in this test have been met.

4.3 Structural Model Assessment

A structural model was developed to examine the relationships between latent constructs in terms of the direction and strength of path coefficients. The structural model evaluation in this study was measured using three testing steps, namely collinearity ($VIF < 5$), path coefficient significance < 0.05 , and model power assessment (R^2) [30]. This process was carried out through bootstrapping testing using 5,000 subsamples (resamples) with the bias-corrected and accelerated (BCA) method. The BCA method was chosen because it is able to correct bias and take into account the asymmetry of the data distribution, thereby producing more reliable estimates. All test results are accompanied

by a 95% confidence interval (2.5% and 97.5%), which provides an overview of the range of parameter variation based on the bootstrap distribution. The application of this method also strengthens the validity and robustness of the structural model, as the estimation results are not only statistically significant but also stable against random sample variations. The results of testing the structural model can be seen in Table 9, Table 10, Table 11, Table 12, and Figure 3.

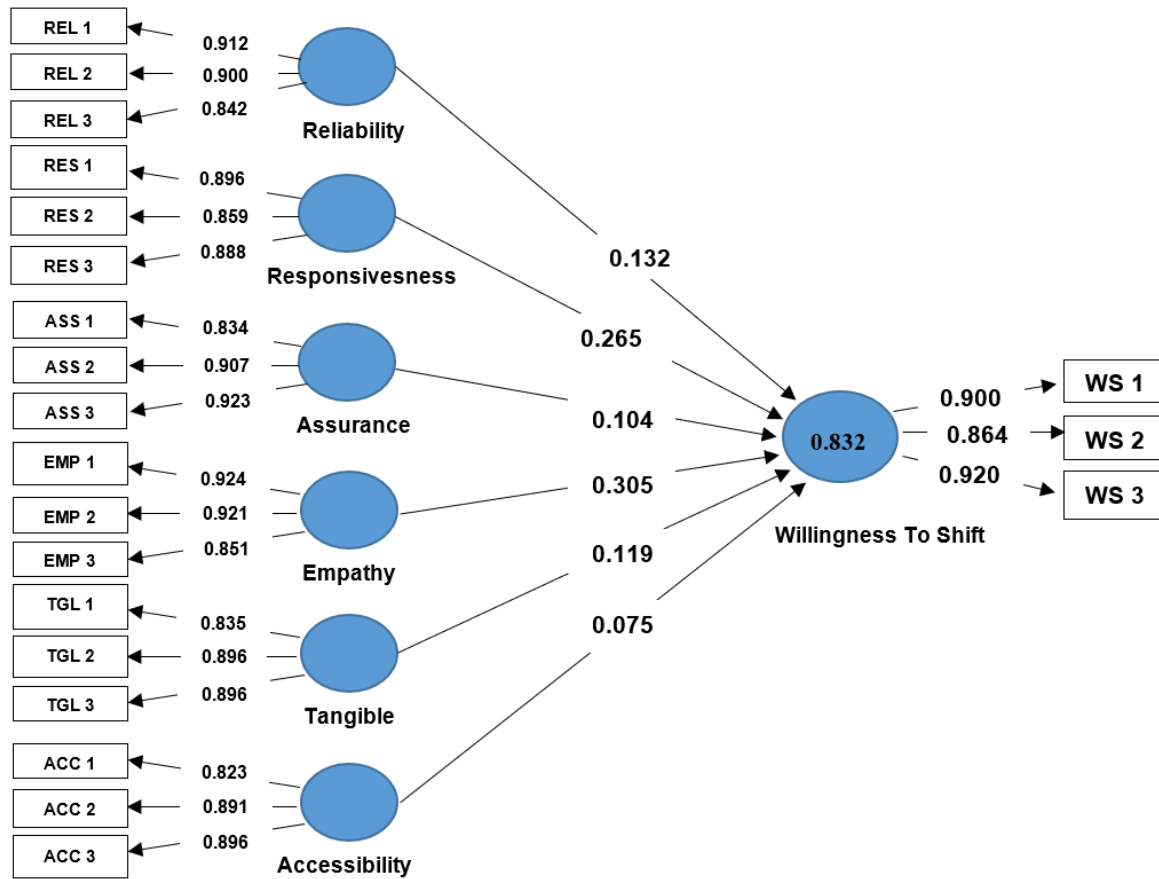


Figure 2. Partial least squares-structural equation modeling (PLS-SEM) algorithm

Meanwhile, the results of the path evaluation in the structural model, as shown in Table 10, indicate that all constructs in the model have a significant effect on the intention to shift with a p -value < 0.05 . However, the level of strength and stability of the influence between constructs differs substantially. Based on the standardised coefficient values, the aspect of empathy ($\beta = 0.305$, $t = 6.556$, $p < 0.000$) emerged as the most dominant factor in encouraging the intention to shift to Bus Trans Jogja, followed by responsiveness ($\beta = 0.265$, $t = 4.446$, $p < 0.000$), reliability ($\beta = 0.132$, $t = 3.659$, $p < 0.000$), tangible ($\beta = 0.119$, $t = 2.551$, $p < 0.011$), assurance ($\beta = 0.104$, $t = 2.606$, $p < 0.009$), and accessibility ($\beta = 0.075$, $t = 2.952$, $p < 0.003$). In terms of significance robustness, the high t -statistics values for empathy, responsiveness, and reliability indicate that these three paths have the most stable influence and are not easily changed by sample variations. Conversely, other constructs such as tangible, assurance, and accessibility have lower t -statistics, indicating that their influence is more sensitive to data changes. These results confirm that the causal strength between constructs in the model is not equal, with empathy, responsiveness, and reliability being the main determinants, while other constructs play a supporting role.

Based on the results of the inner collinearity statistic (VIF) test as shown in Table 9, it can be seen that all variables affecting the intention to shift have values below the threshold of 5, meaning that there is no multicollinearity problem between constructs in the model structure. The highest VIF value is shown by tangible (4.557), followed by responsiveness (4.319), reliability (4.183), assurance (3.934), accessibility (3.620), and Empathy (3.030). These results indicate that the relationship between constructs is still within acceptable limits and does not interfere with path coefficient estimation. Substantively, the higher VIF values for the tangible, responsiveness, and reliability variables indicate that these three variables have more dominant causal power and are closely related in explaining the variation in shifting intention compared to other constructs.

Furthermore, based on the test results as shown in Table 11, it can be seen that the R^2 value is 0.832, indicating that the structural model has a high explanatory power. This means that 83.2% of the variation in the dependent

variable (intention to shift) can be explained by the combination of independent variables in the model (reliability, responsiveness, assurance, empathy, tangibility, and accessibility). The very small difference between the R^2 and adjusted R^2 values confirms that there are no excessive or redundant independent variables in the model. Thus, the model is considered efficient and capable of capturing strong causal relationships between construct variables. Substantively, these results indicate that overall service quality has a significant influence in encouraging people's intention to shift to using the Trans Jogja Bus. After confirming the explanatory power of the model through R^2 , the next step is to conduct a Standardized Root Mean Squared Residual (SRMR) test to evaluate the model fit. Based on the test results as shown in Table 12, it can be seen that the SRMR value is 0.089, which indicates that the model is still within the 'acceptable fit' category and meets the statistical and methodological model feasibility standards [53]. These results indicate that the estimated structural model has good data representation capabilities and does not contain significant deviations in the reproduction of correlations between variables. Therefore, overall, this model can be said to have adequate model fit and can be relied upon for drawing theoretical conclusions and policy implications.

Table 6. Loading, Cronbach's α , composite reliability (ρ_A), composite reliability (ρ_c), and average variance extracted (AVE)

Variable/Instrument	Loadings	Cronbach's α	ρ_A	ρ_c	AVE
Accessibility (ACC)		0.841	0.848	0.903	0.758
ACC1	0.823				
ACC2	0.891				
ACC3	0.896				
Assurance (ASS)		0.867	0.869	0.919	0.790
ASS1	0.834				
ASS2	0.907				
ASS3	0.923				
Empathy (EMP)		0.881	0.884	0.927	0.808
EMP1	0.924				
EMP2	0.921				
EMP3	0.851				
Reliability (REL)		0.861	0.864	0.916	0.784
REL1	0.912				
REL2	0.900				
REL3	0.842				
Responsiveness (RES)		0.857	0.863	0.912	0.776
RES1	0.896				
RES2	0.859				
RES3	0.888				
Tangible (TGL)		0.849	0.855	0.908	0.768
TGL1	0.835				
TGL2	0.896				
TGL3	0.896				
Willingness to shift (WTS)		0.875	0.875	0.923	0.801
WTS1	0.900				
WTS2	0.864				
WTS3	0.920				

Table 7. Discriminant validity–Fornell-Larcker Criterion

Variable	Accessibility	Assurance	Empathy	Reliability	Responsiveness	Tangible	Willingness To Shift
Accessibility	0.870						
Assurance	0.821	0.889					
Empathy	0.710	0.767	0.899				
Reliability	0.788	0.844	0.735	0.885			
Responsiveness	0.763	0.841	0.788	0.795	0.881		
Tangible	0.781	0.843	0.737	0.789	0.859	0.876	
Willingness to shift	0.777	0.835	0.833	0.809	0.858	0.823	0.895

After obtaining the R^2 and SRMR values, a PLSpredict analysis was conducted to reinforce and validate the results. The PLSpredict test results are presented in Table 13.

Based on the results of the PLSpredict analysis as shown in Table 13, it can be seen that the model has excellent predictive capabilities. Q^2 values above 0 (WS1 = 0.634, WS2 = 0.683, and WS3 = 0.663) indicate that the model has strong predictive relevance. Furthermore, the PLS-SEM RMSE and PLS-SEM MAE values are smaller than the linear regression (LM)-SEM RMSE and LM-SEM MAE values, indicating that the PLS-SEM model produces lower prediction errors than the traditional LM model. Thus, the PLSpredict results confirm that the model not only has high explanatory power (through R^2) but also has superior predictive capabilities both technically and empirically. Therefore, the model can be declared robust and valid both in explaining the phenomenon of shifting intentions and in predicting future user behaviour.

Table 8. Discriminant validity–cross loading

Instrument	Accessibility (ACC)	Assurance (ASS)	Empathy (EMP)	Reliability (REL)	Responsiveness (RES)	Tangible (TGL)	Willingness To Shift (WTS)
ACC1	0.823	0.747	0.606	0.678	0.713	0.693	0.771
ACC2	0.891	0.688	0.624	0.682	0.630	0.665	0.616
ACC3	0.896	0.687	0.616	0.688	0.624	0.668	0.606
ASS1	0.738	0.834	0.638	0.713	0.760	0.740	0.810
ASS2	0.716	0.907	0.685	0.764	0.723	0.742	0.680
ASS3	0.724	0.923	0.720	0.770	0.747	0.759	0.716
EMP1	0.640	0.686	0.924	0.674	0.718	0.671	0.695
EMP2	0.621	0.692	0.921	0.663	0.706	0.658	0.713
EMP3	0.645	0.685	0.851	0.642	0.696	0.654	0.818
REL1	0.696	0.726	0.682	0.912	0.699	0.679	0.721
REL2	0.711	0.786	0.633	0.900	0.693	0.702	0.744
REL3	0.686	0.727	0.638	0.842	0.722	0.717	0.682
RES1	0.676	0.759	0.744	0.716	0.896	0.782	0.732
RES2	0.696	0.756	0.660	0.682	0.859	0.780	0.838
RES3	0.635	0.699	0.679	0.705	0.888	0.696	0.679
TGL1	0.710	0.766	0.624	0.685	0.802	0.835	0.812
TGL2	0.652	0.703	0.640	0.679	0.705	0.896	0.656
TGL3	0.677	0.732	0.670	0.704	0.729	0.896	0.666
WTS1	0.673	0.750	0.714	0.712	0.760	0.731	0.900
WTS2	0.700	0.762	0.763	0.739	0.784	0.728	0.864
WTS3	0.711	0.729	0.757	0.720	0.758	0.750	0.920

Table 9. Inner model collinearity statistic (VIF)

Path	Original Sample (O)	Sample Mean (M)	2.5%	97.5%
Accessibility → Willingness to shift	3.620	3.701	2.889	4.832
Assurance → Willingness to shift	3.934	4.058	3.792	4.549
Empathy → Willingness to shift	3.030	3.108	2.506	3.906
Reliability → Willingness to shift	4.183	4.255	3.531	4.421
Responsiveness → Willingness to shift	4.319	4.463	4.061	4.706
Tangible → Willingness to shift	4.557	4.783	4.011	4.947

Table 10. Path coefficient—mean, standard deviation (STDEV), *t*-values, *p*-values (hypotheses testing)

Hypotheses	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T-Statistics (O/STDEV)	P-Values
Accessibility → Willingness to shift	0.075	0.077	0.025	2.952	0.003
Assurance → Willingness to shift	0.104	0.105	0.040	2.606	0.009
Empathy → Willingness to shift	0.305	0.305	0.047	6.556	0.000
Reliability → Willingness to shift	0.132	0.133	0.036	3.659	0.000
Responsiveness → Willingness to shift	0.265	0.262	0.060	4.446	0.000
Tangible → Willingness to shift	0.119	0.119	0.047	2.551	0.011

Table 11. Model power assessment (R^2)

Variable	R^2	R^2 Adjusted
Willingness to shift	0.832	0.831

Table 12. Standardized Root Mean Square Residual (SRMR)

Model Fit Index	Saturated Model	Estimated Model
SRMR	0.089	0.089

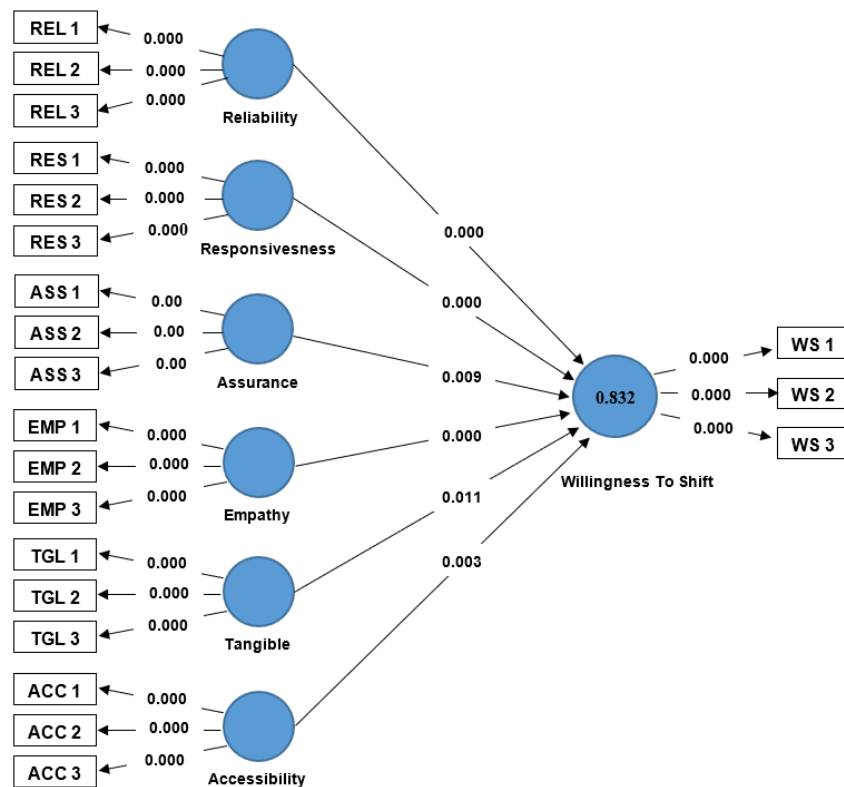
**Figure 3.** Bootstrapping graphic

Table 13. Discriminant validity–cross loading

Item	Q^2 predict	PLS-SEM_RMSE	PLS-SEM_MAE	LM_RMSE	LM_MAE
WS1	0.634	0.505	0.326	0.545	0.366
WS2	0.683	0.398	0.298	0.454	0.342
WS3	0.663	0.402	0.257	0.411	0.277

Note: PLS-SEM_RMSE: partial least squares-structural equation modeling_root mean square error; PLS-SEM_MAE: partial least squares-structural equation modeling_mean absolute error; LM_RMSE: linear model_root mean square error; LM_MAE: linear model_mean absolute error; WS: willing to shift.

5 Discussion

Reliability factors, which include waiting times, predictable travel times, and consistent bus arrival frequencies, are key aspects that greatly influence people's decisions to shift to public transport. In mode choice logic, users tend to choose modes that provide certainty and minimise discomfort due to schedule uncertainty. This is in line with a review stating that frequency, reliability, and waiting times can increase the desire to use public transport [54]. This is also reinforced by findings from research stating that improving the reliability of bus travel times can attract more passengers to use buses, thereby reducing the number of cars [54].

Responsiveness, which reflects the readiness of officers to respond to passengers' needs and complaints, is an important factor in increasing the trust and comfort of public transport users. When staff are able to provide information quickly, assist during disruptions, and show concern, perceptions of service quality will improve and encourage people to shift from private vehicles to public transport. This is in line with findings in Madrid, which show that service quality, satisfaction and attitudes towards public transport are considered the main motivating factors behind public transport users' behaviour [55]. In addition, other studies confirm that staff behaviour as part of responsiveness plays an important role in shaping passenger satisfaction [56]. Thus, improving the responsiveness factor of Trans Jogja will strengthen the public's intention to shift to using it.

Improvements in assurance dimensions such as service guarantees, sufficient seating and standing room on buses, and comfort during the journey can significantly encourage passengers to shift to bus services. This is in line with a meta review study which states that comfort, such as getting a seat, is very important in mode of transport preference [12].

The empathy factor of public transport officers, such as providing clear information, personalised service, and the friendliness of officers on the bus, can directly reduce the psychological burden of travelling, such as confusion, anxiety, and even feelings of being unappreciated during the journey. Such attitudes are highly appreciated by passengers because they feel valued, safe, and comfortable when using public transport such as buses. This is supported by research findings stating that customer service quality, such as the attitude of staff, is one of the main drivers influencing preferences and satisfaction, which can encourage users to shift to using public transport such as buses [57].

Tangible aspects related to bus services, such as the number and location of strategic bus stops, dedicated bus lanes, and parking facilities, have proven to be important factors that can encourage people to shift from private vehicles to buses. Bus stops that are evenly distributed and close to centres of activity can facilitate access, while dedicated bus lanes can increase travel speed and punctuality, making buses more competitive than private vehicles. Bus travel times along exclusive bus lanes were found to be significantly higher than other traffic (especially cars) along adjacent traffic lanes [58]. In addition, park-and-ride facilities near bus stops allow private vehicles to combine their mobility with buses, which can reduce first-mile barriers to bus stops. This is also confirmed by the results of a study which states that the availability of park-and-ride facilities near bus stops can increase the number of passengers. In the context of buses, this means that integrating private vehicle access with bus stops can strengthen public interest in shifting to public transport [59].

Accessibility, such as ease of purchasing tickets, extensive route coverage, and ease of direct transfers between modes of transport, reduces technical and psychological barriers to choosing buses, thereby increasing the desire to use public transport such as buses. With an easy ticketing system, for example using integrated cards or applications, passengers are no longer burdened with purchasing tickets. Good route coverage and smooth transfers can reduce travel time. This combination makes buses more competitive than private vehicles. This is in line with research showing that attributes such as fare integration, ease of access, and good route networks can be determining factors in the attractiveness of public transport, which can increase the intention of potential passengers to use public transport [12, 60].

Overall, the results of this study indicate that the quality of public transport services, including reliability, responsiveness, assurance, empathy, and accessibility, plays an important role in increasing the public's desire to shift to public transport such as the Trans Jogja Bus. The high level of reliability and responsiveness of operators

creates a sense of confidence that the journey will run on time as expected, while the assurance of safety and professionalism of officers can increase passenger safety. On the other hand, friendly treatment and attention to individual needs (empathy) increase emotional comfort. Good accessibility in terms of ease of purchasing tickets, route coverage, and ease of transfer can reduce practical barriers to using buses. The combination of these factors makes public transport more competitive than private vehicles, thereby encouraging an increase in the public's desire to shift to using the Trans Jogja Bus on a sustainable basis.

6 Limitation and Future Research

Although the results of this study provide an important contribution to understanding the potential for shifting to the Trans Jogja Bus, there are still several limitations to this study. These limitations include the following:

1) The SEM model used is cross-sectional, so that the relationships between variables reflect respondents' perceptions at a specific point in time and cannot fully describe actual changes in transport user behaviour over time. Furthermore, this model relies on perception data from questionnaires, so the results obtained are highly dependent on the subjectivity and honesty of respondents in providing answers.

2) There is a possibility of social desirability bias, which may cause respondents to give answers that are considered socially acceptable, such as expressing a greater desire to use public transport for environmental or moral reasons. To minimise this bias, all respondents were assured of confidentiality in providing their answers.

3) The potential for self selection bias is also a limitation of this study. Although sampling was conducted randomly at various activity centres along the Trans Jogja bus route, there is still a possibility that individuals who completed the questionnaire had a particular interest in public transport, which could influence the results of the study. Therefore, to reduce this bias, the questionnaire was distributed widely in various locations and activity segments (education, health, offices, tourist destinations, shopping centres, and government agencies). Thus, it is hoped that a more diverse representation of the population of private vehicle and online vehicle users who are the sample in this study can be obtained.

Based on this, it is recommended that further research combine data on perceptions with actual user behaviour, for example through travel data (GPS tracking) or longitudinal surveys, and expand the scope of the research sample using randomised or stratified sampling methods in order to increase external validity and reduce bias in the model.

7 Conclusion

The results of this study indicate that a person's decision to shift to using the Trans Jogja Bus is greatly influenced by factors directly related to comfort, affordability, and travel efficiency. Indicators such as lower prices compared to private vehicles, short waiting times, and the availability of supporting facilities have been proven to play an important role in encouraging people to shift. The logic of choosing public transport confirms that when bus services can offer lower travel costs, competitive travel times, and adequate comfort, the likelihood of shifting to buses will be higher.

In addition, service reliability aspects, including punctuality, frequency of arrival, and consistency of service quality are crucial factors in encouraging people to shift providers. Users will prefer public transport if they can trust the system to meet their daily mobility needs without the risk of significant delays. This is in line with mode choice theory, whereby users tend to choose alternatives with lower generalised travel costs, which is a combination of monetary costs, time and convenience.

The findings of this study also have important implications for the development of public transport systems in Yogyakarta. The local government and Trans Jogja bus operators need to prioritise policies that improve service quality across the board. This is particularly important in terms of punctuality, comfort and accessibility of bus stops. These efforts can be achieved by increasing the number of buses, implementing consistent departure schedules, and providing supporting facilities such as shaded waiting areas and real time bus arrival information.

In terms of fare policy, a competitive and affordable pricing scheme is needed for the public, for example through the integration of digital payment systems and fare discounts for regular users or students. The government is also advised to strengthen coordination between agencies in providing priority bus lanes so that travel times become more efficient and attractive to private vehicle users.

Furthermore, long term policies need to be directed towards the development of an integrated transport network that connects bus services with feeder modes (such as environmentally friendly vehicles or park and ride facilities). With policy strategies that focus on quality, efficiency and affordability, the potential for the public to shift to Bus Trans Jogja can increase significantly. In addition, this can support efforts to reduce congestion and emissions in the urban area of Yogyakarta.

Author Contributions

Conceptualization, G.R.P., S.P., and M.Z.I.; methodology, G.R.P., S.P., and M.Z.I.; investigation, G.R.P.; formal analysis, G.R.P., S.P., and M.Z.I.; data curation, G.R.P.; writing—original draft preparation, G.R.P.; writing—review

and editing, S.P. and M.Z.I.; visualization, G.R.P.; validation, G.R.P., S.P., and M.Z.I.; resources, G.R.P., S.P., and M.Z.I.; supervision, S.P. and M.Z.I.; project administration, G.R.P.; funding acquisition, G.R.P. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

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